Sri Eshwar College of Engineering

(An Autonomous Institution)

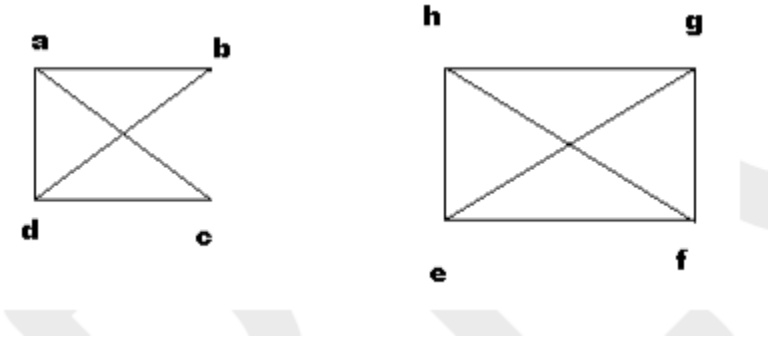
Approved by AICTE, New Delhi and Affiliated to Anna University, Chennai

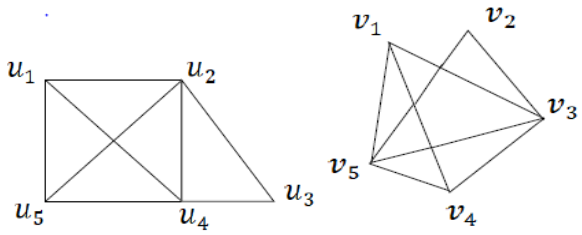
Kondampatti (Post), Kinathukadavu (Tk), Coimbatore - 641 202

U19MA204 – DISCRETE MATHEMATICS

TUTORIAL-MODULE IV – GRAPH THEORY

CO4 : Analyze the types of circuits in a graph, the existence of isomorphism and sketch the Euler and Hamiltonian paths and circuits in a graph.

	TUTORIAL-I		
S. No	**QUESTIONS**	**CO**	**BT**
1.	Prove that the maximum number of edges in simple disconnected graph G with n vertices and k components is $\frac{(n-k)(n-k+1)}{2}$.	4	K3
2	Establish the isomorphism between the graph. 	4	K3
3	Prove that the number of edges in a bipartite graph with n vertices is at most $\frac{n^2}{2}$	4	K3
	TUTORIALI-II		
S. No	**QUESTIONS**	**CO**	**BT**
1.	Show that graph G is disconnected if and only if its vertex set V can be partitioned into two non empty subset s V_1 and V_2 such that there exists no edge in G whose one end vertex is in V_1 and the other in V_2.	4	K3
2	If a graph G (either connected or not) has exactly two vertices of odd degree there is a path joining these two vertices.	4	K3

	TUTORIALI-II		
S. No	**QUESTIONS**	**CO**	**BT**
1.	Using Python, Prove that if a graph G has not more than two vertices of odd degree, then there can be Euler path in G.	4	K3
2	Prove that a connected graph G is Eulerian iff all the vertices are of even degree.	4	K3
3	Check whether the following two graphs given below are isomorphic or not using circuit method. 	4	K3

CO	PO 01	PO 02	PO 03	PO 04	PO 05	PO 06	PO 07	PO 08	PO 09	PO 10	PO 11	PSO 01	PSO 02
U23MA204.4	3	3	2	2	2	1	-	-	-	-	2	-	-

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